

79
100

Hugo Guerrero

9/1/19 che 210

HW Set 1

#1) Calculate the coefficient of determination R^2 as a function of the angle θ for a pure rotation. Use the data set & plot the results to determine the maximum & minimum values of the function.

Pure Rotation: $\hat{X}_i = [X_i \cos(\theta) - Y_i \sin(\theta)]$ & $\hat{Y}_i = [X_i \sin(\theta) + Y_i \cos(\theta)]$

$$R^2 = \frac{(\sum \hat{X}_i \hat{Y}_i)^2}{(\sum \hat{X}_i^2)(\sum \hat{Y}_i^2)} \quad \text{where} \quad \begin{aligned} \sum \hat{X}_i \hat{Y}_i &= \sum X_i Y_i - \sum X_i^2 \sin(2\theta) \\ \sum \hat{X}_i^2 &= \sum X_i^2 \cos^2(\theta) + \sum Y_i^2 \sin^2(\theta) - 2 \sum X_i Y_i \sin(2\theta) \\ \sum \hat{Y}_i^2 &= \sum X_i^2 \sin^2(\theta) + \sum Y_i^2 \cos^2(\theta) + 2 \sum X_i Y_i \sin(2\theta) \end{aligned} \quad +4$$

$$\bullet \sum \hat{X}_i = \frac{1}{n} \sum \hat{X}_i = \frac{1}{n} \sum (X_i \cos(\theta) - Y_i \sin(\theta)) = \left(\frac{1}{n} \sum X_i \right) \cos(\theta) - \left(\frac{1}{n} \sum Y_i \right) \sin(\theta) = \bar{X} \cos(\theta) - \bar{Y} \sin(\theta) \quad \checkmark$$

$$\bullet \sum \hat{X}_i^2 = \frac{1}{n} \sum X_i^2 = \frac{1}{2} \sum (X_i \cos(\theta) - Y_i \sin(\theta))^2 = \sum X_i^2 \cos^2(\theta) + \sum Y_i^2 \sin^2(\theta) - 2 \sum X_i Y_i \sin(2\theta) \quad \checkmark$$

$$\bullet \sum \hat{Y}_i = \frac{1}{n} \sum \hat{Y}_i = \frac{1}{n} \sum (X_i \sin(\theta) + Y_i \cos(\theta)) = \left(\frac{1}{n} \sum X_i \right) \sin(\theta) + \left(\frac{1}{n} \sum Y_i \right) \cos(\theta) = \bar{X} \sin(\theta) + \bar{Y} \cos(\theta) \quad \checkmark +2$$

$$\bullet \sum \hat{Y}_i^2 = \frac{1}{n} \sum Y_i^2 = \frac{1}{2} \sum (X_i \sin(\theta) + Y_i \cos(\theta))^2 = \sum X_i^2 \sin^2(\theta) + \sum Y_i^2 \cos^2(\theta) + 2 \sum X_i Y_i \sin(2\theta) \quad \checkmark +4$$

$$\bullet \sum \hat{X}_i \hat{Y}_i = \frac{1}{n} \sum \hat{X}_i \hat{Y}_i = \frac{1}{n} \sum [(X_i \cos(\theta) - Y_i \sin(\theta)) \cdot (X_i \sin(\theta) + Y_i \cos(\theta))] = \sum X_i^2 \cos(\theta) \sin(\theta) + \sum X_i Y_i \cos^2(\theta) - \sum X_i Y_i \sin^2(\theta) - \sum Y_i^2 \cos(\theta) \sin(\theta) \quad \checkmark$$

$$R^2 = \frac{[\sum X_i^2 \cos(\theta) \sin(\theta) + \sum X_i Y_i \cos^2(\theta) - \sum X_i Y_i \sin^2(\theta) - \sum Y_i^2 \cos(\theta) \sin(\theta)]^2}{[\sum X_i^2 \cos^2(\theta) + \sum Y_i^2 \sin^2(\theta) - 2 \sum X_i Y_i \sin(2\theta)] [\sum X_i^2 \sin^2(\theta) + \sum Y_i^2 \cos^2(\theta) + 2 \sum X_i Y_i \sin(2\theta)]}$$

$$R^2 = \frac{[\sum X_i Y_i \cos^2(\theta) - \sum X_i Y_i \sin^2(\theta)]^2}{[\sum X_i^2 \cos^2(\theta) + \sum Y_i^2 \sin^2(\theta) - 2 \sum X_i Y_i \sin(2\theta)] [\sum X_i^2 \sin^2(\theta) + \sum Y_i^2 \cos^2(\theta) + 2 \sum X_i Y_i \sin(2\theta)]}$$

$$R^2 = \frac{[\sum X_i Y_i \cos^2(\theta) - \sum X_i Y_i \sin^2(\theta)]^2}{[\sum X_i^2 \cos^2(\theta) + \sum Y_i^2 \sin^2(\theta) - 2 \sum X_i Y_i \sin(2\theta)] [\sum X_i^2 \sin^2(\theta) + \sum Y_i^2 \cos^2(\theta) + 2 \sum X_i Y_i \sin(2\theta)]}$$

I don't believe these cancel like that.

+10

$$r = \frac{S_{xy} - S_x S_y}{S_{xx} - S_x^2}$$

$$= 2.019$$

Hugo Guerrero

9/1/19 CHE 210

Pt. 2

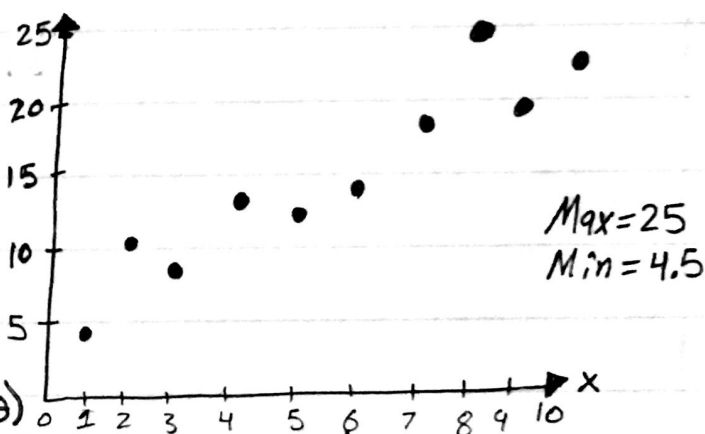
$$\theta = \arctan(\text{slope for best fit line}) = 63.65$$

X_i	Y_i	X_i^2	$X_i Y_i$	Y_i^2
1	4.5	1	4.5	20.25
2	11	4	22	121
3	8	9	24	64
4	13.9	16	55.6	193.21
5	12	25	60	144
6	14	36	84	196
7	18	49	126	324
8	25	64	200	625
9	19	81	171	361
10	23.6	100	239	571.21
5.5	14.9	38.5	98.61	261.967

$$R^2 = \frac{(SS_{xy})^2}{(SS_{xx})(SS_{yy})} = \frac{(S_{xy} - S_x S_y)^2}{(S_{xx} - S_x^2)(S_{yy} - S_y^2)}$$

$$R^2 = \frac{(98.61 - [5.5 \cdot 14.9])^2}{([5.5]^2 - 38.5)([14.9]^2 - 261.967)}$$

$$R^2 = \frac{277.5556}{329.64525} = \boxed{0.8419}$$



Need to graph $R^2(\theta)$ (is a function of theta) that you derived on page 1.

#2) Derive the least squares method for a linear equation, $Y = ax + b$, using horizontal offsets. Find expressions for the two coefficients in terms of sums defined as S_x, S_y, S_{xx} , etc. Fit the data for both vertical & horizontal offsets, why do you think they are different?

$$Y = ax + b, \quad d_i = X_i - X(Y_i) = X_i - \left(\frac{Y_i - b}{a}\right) + 1$$

$$X = \frac{Y - b}{a} \quad \phi(a, b) = \sum_{i=1}^n d_i^2 = \sum_{i=1}^n \left[X_i - \left(\frac{Y_i - b}{a}\right) \right]^2$$

$$d_i = X_i - \left(\frac{Y_i - b}{a}\right)$$

$$\frac{\partial \phi}{\partial a} = 0 = 2 \sum_{i=1}^n \left[X_i - \left(\frac{Y_i - b}{a}\right) \right] \cdot \left(-\frac{Y_i - b}{a^2}\right)$$

$$\frac{\partial \phi}{\partial a} = 0 = 2 \sum_{i=1}^n [X_i - (ay_i + b)] \cdot (-y_i)$$

$$\frac{\partial \phi}{\partial b} = 0 = 2 \sum_{i=1}^n \left[X_i - \left(\frac{Y_i - b}{a}\right) \right] \cdot \left(-\frac{1}{a}\right)$$

$$\frac{\partial \phi}{\partial b} = 0 = 2 \sum_{i=1}^n [X_i - (ay_i + b)] \cdot (-1)$$

$$\left[\sum_{i=1}^n (Y_i - b)^2 \right] = \frac{\sum_{i=1}^n Y_i^2 - \sum_{i=1}^n Y_i b}{a^2}$$

$$[a \sum_{i=1}^n Y_i^2 + b \sum_{i=1}^n Y_i = \sum_{i=1}^n X_i Y_i]$$

$$[a \sum_{i=1}^n Y_i + b n = \sum_{i=1}^n X_i]$$

$$\left[\frac{\sum_{i=1}^n Y_i - b}{a^2} = \frac{\sum_{i=1}^n X_i}{a} \right]$$

$$\left[\frac{(S_y - b)^2}{a} = S_{xy} - S_x b \right]$$

$$\left[\frac{S_y - b}{a} = S_x \right]$$

$$[S_y a + S_y b = S_{xy}]$$

$$[S_y a + b = S_x] \quad +12$$

Hugo Guerrero

9/1/19 CHE 210

Pt. 3

$$\begin{bmatrix} S_{yy} & -2S_y & S_y \\ -2S_y & S_x & -1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} S_{xy} \\ S_x \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} S_{yy} & S_y \\ S_y & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} S_{xy} \\ S_x \end{bmatrix}$$

$$\det(A) = S_{yy} - S_y^2$$

$$\det(A_1) = \det \begin{bmatrix} S_{xy} & S_y \\ S_x & 1 \end{bmatrix} = S_{xy} - S_y S_x$$

$$\det(A_2) = \det \begin{bmatrix} S_{yy} & S_{xy} \\ S_y & S_x \end{bmatrix} = S_{yy} S_x - S_{xy} S_y$$

$$a = \frac{\det(A_1)}{\det(A)} = \frac{S_{xy} - S_y S_x}{S_{yy} - S_y^2}$$

$$b = \frac{\det(A_2)}{\det(A)} = \frac{S_{yy} S_x - S_{xy} S_y}{S_{yy} - S_y^2}$$

$$a = \frac{(98.61) - (5.5 \cdot 14.9)}{(14.9)^2 - (261.967)} = \frac{16.66}{-39.957} = \boxed{-0.42}$$

$$b = \frac{(14.9)^2(5.5) - (98.61)(14.9)}{(14.9)^2 - (261.967)} = \frac{-248.234}{-39.957} = \boxed{6.21}$$

$$Y = (-0.42)X + (6.21)$$

Murphy

1.2

#3) Nitroglycerin $[C_3H_5(NO_3)_3]$ is an explosive that decomposes to CO_2 , H_2O , N_2 , and O_2 . Write a balanced chemical equation.



$$C: 3V_{C_3H_5(NO_3)_3} = V_{CO_2} + 0 + 0 + 0$$

$$H: 5V_{C_3H_5(NO_3)_3} = 0 + 2V_{H_2O} + 0 + 0$$

$$N: 3V_{C_3H_5(NO_3)_3} = 0 + 0 + 2V_{N_2} + 0$$

$$O: 9V_{C_3H_5(NO_3)_3} = 2V_{CO_2} + V_{H_2O} + 0 + 2V_{O_2}$$

$$\begin{bmatrix} C \\ H \\ N \\ O \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 2 & 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} V_{CO_2} \\ V_{H_2O} \\ V_{N_2} \\ V_{O_2} \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \\ 3 \\ 9 \end{bmatrix}$$

$$\det(A) = 8 \quad \det(A_3) = 12$$

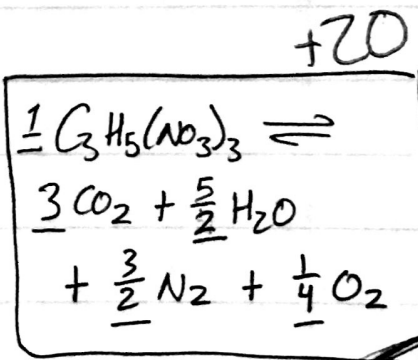
$$\det(A_1) = 24 \quad \det(A_4) = 2$$

$$\det(A_2) = 20$$

$$V_{CO_2} = \frac{24}{8} = \boxed{3} \quad V_{C_3H_5(NO_3)_3} = \frac{8}{8} = \boxed{1}$$

$$V_{H_2O} = \frac{20}{8} = \boxed{\frac{5}{2}} \quad V_{O_2} = \frac{2}{8} = \boxed{\frac{1}{4}}$$

$$V_{N_2} = \frac{12}{8} = \boxed{\frac{3}{2}}$$



+20

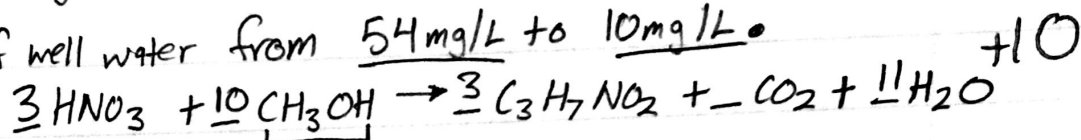
Hugo Guerrero

9/1/19 CHE 210

pt. 4

Murphy 1.15

#4) The equation as written is not balanced. Find the correct stoichiometry coefficients. Calculate the grams of methanol required to reduce the nitrate content (as HNO_3) of 10L of well water from 54 mg/L to 10 mg/L .



$$\text{C: } 0 - V_{\text{CH}_3\text{OH}} + 3V_{\text{C}_3\text{H}_7\text{NO}_2} + V_{\text{CO}_2} = 0$$

$$\text{H: } -V_{\text{HNO}_3} - 4V_{\text{CH}_3\text{OH}} + 7V_{\text{C}_3\text{H}_7\text{NO}_2} + 0 + 2V_{\text{H}_2\text{O}} = 0$$

$$\text{N: } -V_{\text{HNO}_3} + 0 + V_{\text{C}_3\text{H}_7\text{NO}_2} + 0 + 0 = 0 \quad V_{\text{HNO}_3} = 1 \quad V_{\text{C}_3\text{H}_7\text{NO}_2} = 1$$

$$\text{O: } -3V_{\text{HNO}_3} - V_{\text{CH}_3\text{OH}} + 2V_{\text{C}_3\text{H}_7\text{NO}_2} + V_{\text{CO}_2} + V_{\text{H}_2\text{O}} = 0$$

$$\bullet \frac{54 \text{ mg HNO}_3}{1 \text{ L}} \left| \frac{10 \text{ L}}{1 \text{ mg}} \right| 10^{-3} = 0.54 \text{ g HNO}_3 \quad \frac{10 \text{ mg HNO}_3}{1 \text{ L}} \left| \frac{10 \text{ L}}{1 \text{ mg}} \right| 10^{-3} = 0.1 \text{ g HNO}_3 + 10$$

$$\bullet 0.54 \text{ g} - 0.1 \text{ g} = 0.44 \text{ g HNO}_3 \left| \frac{1 \text{ mol HNO}_3}{63 \text{ g HNO}_3} \right| \frac{10 \text{ mol CH}_3\text{OH}}{3 \text{ mol HNO}_3} \left| \frac{32 \text{ g CH}_3\text{OH}}{1 \text{ mol CH}_3\text{OH}} \right| = 0.74 \text{ g CH}_3\text{OH}$$

$$\bullet 1 \text{ mol HNO}_3 = 63 \text{ g HNO}_3$$

$$\bullet 1 \text{ mol CH}_3\text{OH} = 32 \text{ g CH}_3\text{OH}$$

Murphy 1.17

#5) What is the wt % Hydrogen in common liquids & solids such as Ethanol ($\text{C}_2\text{H}_5\text{OH}$), methane (CH_4), water (H_2O), and glucose ($\text{C}_6\text{H}_{12}\text{O}_6$)?

$$\bullet \text{Ethanol mass} = 46 \text{ g} \quad \frac{6}{46} \times 100 = 13.04\% \quad \checkmark \quad \bullet \text{Methane mass} = 16 \text{ g} \quad \frac{4}{16} \times 100 = 25 \text{ wt}\% \quad \checkmark$$

$\text{H mass} = 6 \text{ g} \quad \text{H mass} = 4 \text{ g}$

$$\bullet \text{Water mass} = 18 \text{ g} \quad \frac{2}{18} \times 100 = 11.11 \text{ wt}\% \quad \checkmark \quad \bullet \text{Glucose mass} = 180 \text{ g} \quad \frac{12}{180} \times 100 = 6.67 \text{ wt}\% \quad \checkmark$$

$\text{H mass} = 2 \text{ g} \quad \text{H mass} = 12 \text{ g}$

- It is right to be skeptical of 50% wt of Hydrogen, when most compounds hold within the 1% to 8% range. +17

+37